UNIVERSITÉ TOULOUSE CAPITOLE

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REPORT

Software Design

Assignment 1

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Chapter 1

Usecase Digrams

1.1 Indicate usecases and actors

From the description of the application, we defined the main actors of our application.

Actors	**Reason**
Customer	As mentioned in the assignment: "...customer starts a reservation by mentioning where they want to pick up the car..."
Agency Employee	As mentioned in the assingment: "Based on the chosen category, the agency employee will allocate a specific car to the customer."
Hotline Operator	As metioned in the assignment: "The 24/7 hotline will contact her back and organize with her the repairing actions"

Table 1.1: Actors in system

Design choices: Each represent a role interacting with the system. We choosed to design the system with one Customer for the actors before and after signing in because the action of authentication change the customer's access to the system, not changing their role. Agency Employee represent the employee working at the agency location and handle pickup and return action. Hotline operator handle report and coordinate assistances following an incident.

These actors with participate in several usecase:

Usecase	**Participation**
Make reservation	Customer
Pay	Customer
Cancel reservation	Customer
Pickup car	Customer, Agency Employee
Return car	Customer, Agency Employee
Manage incident	Customer, Hotline Operator
Report incident	Customer

Table 1.2: List of usecases and their participations

1.2 Usecase diagram

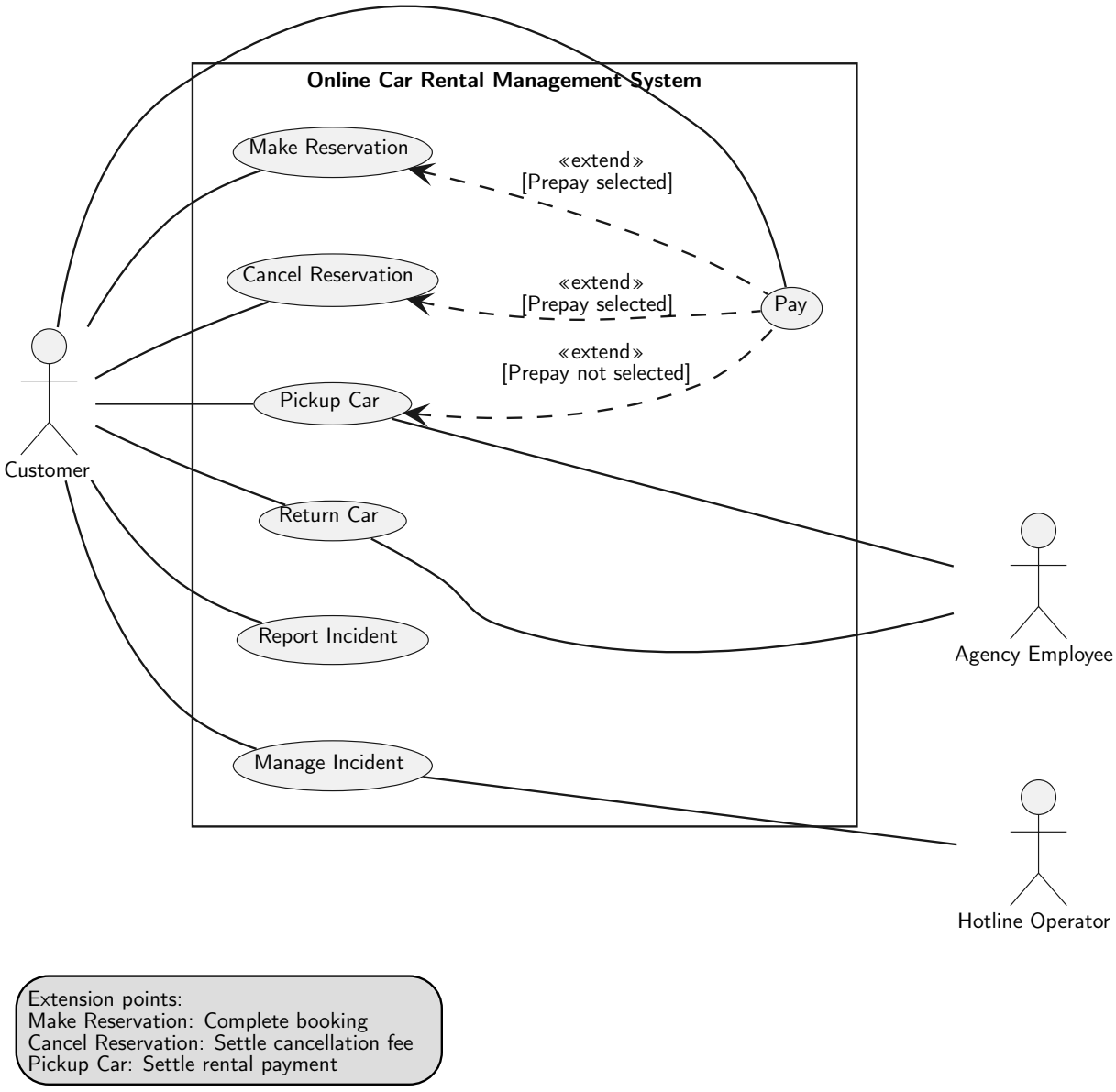


Figure 1.1: General usecase diagram

1.3 Detail of use cases associated to the action "pick up car"

1.3.1 System sequence diagram

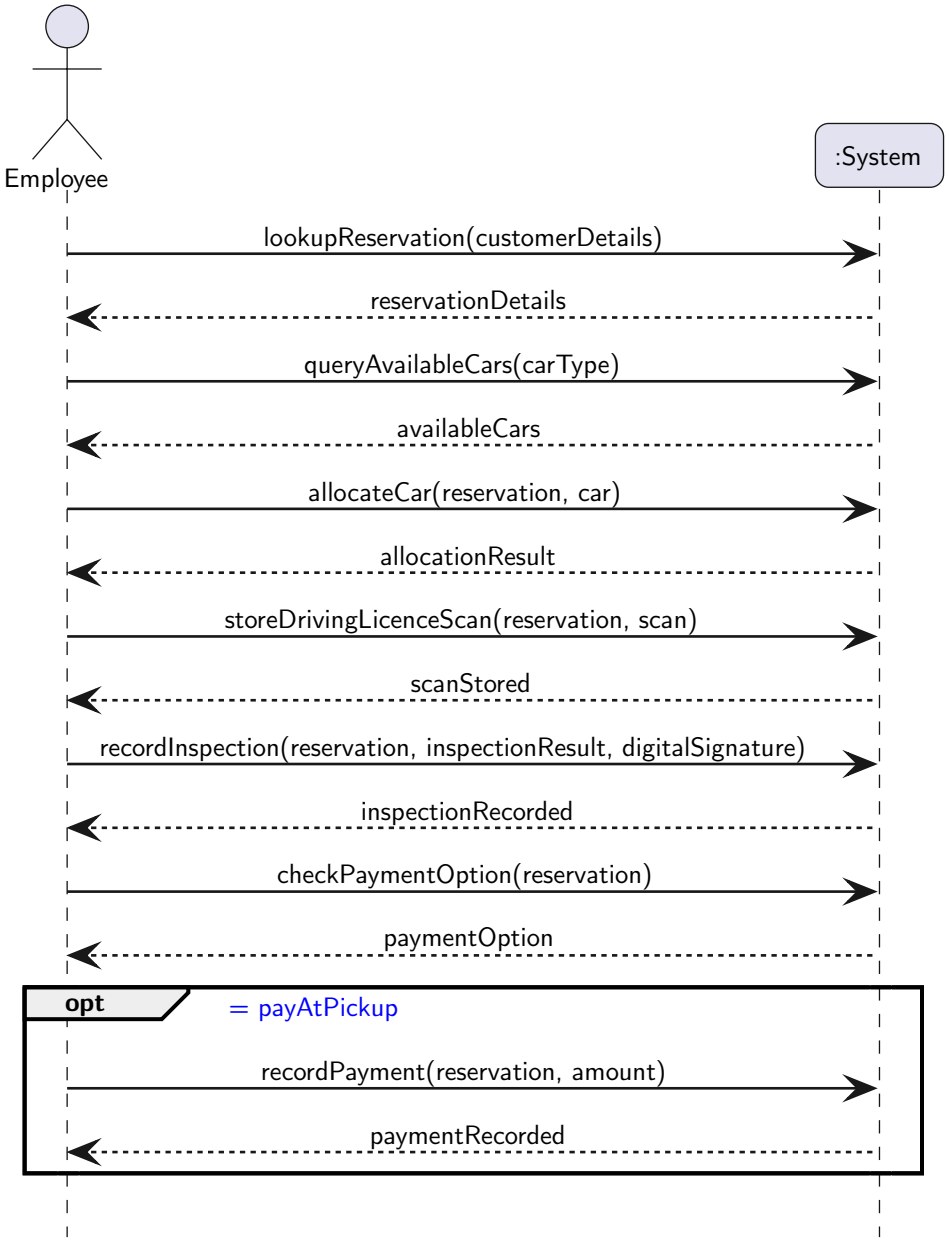


Figure 1.2: System sequence diagram for pickup car action

1.3.2 Lo-Fi interface

We have designed a lo-fi interface serving the pickup action. The activities follow the sequence: find reservation, review it details, allocate a car from the booked category, check and scan the driver license, record inspection result with customer's digital signature.

You can click on the navigation button link the the corresponding figures in this PDF

that help illustrate the workflow steps. Every actions must succeed before going to the next step and the key is only handed out after all checks and payment are complete.

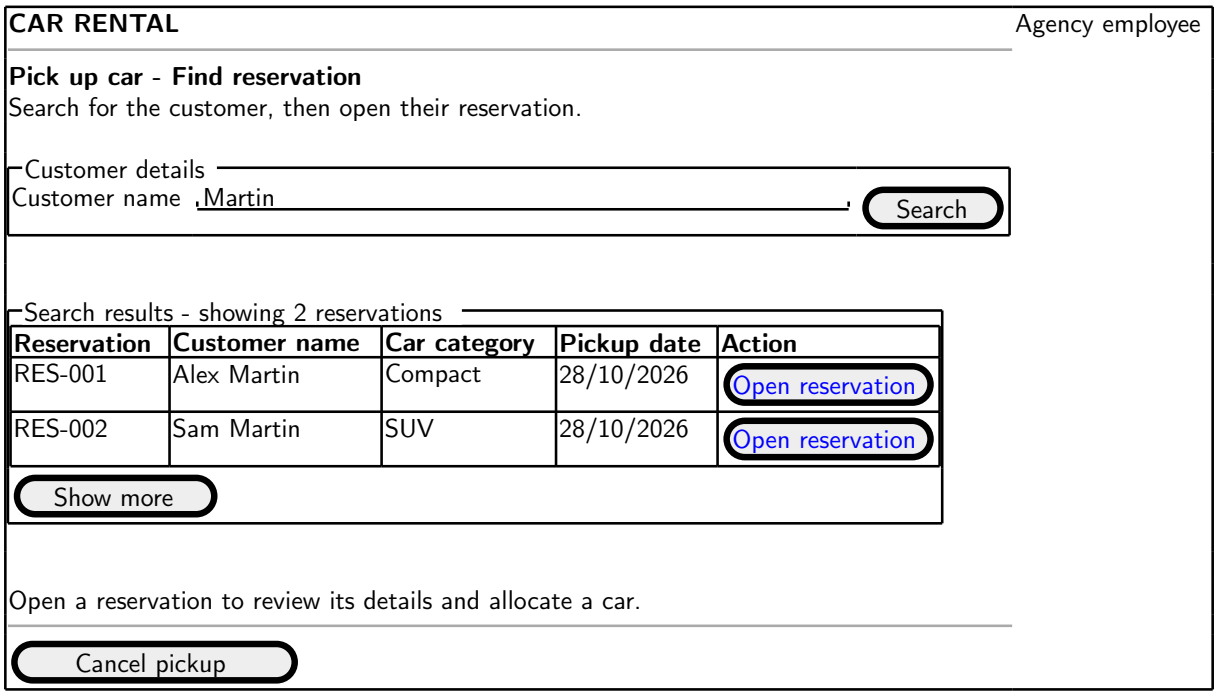


Figure 1.3: Main screen of Lo-fi inteface for pickup action

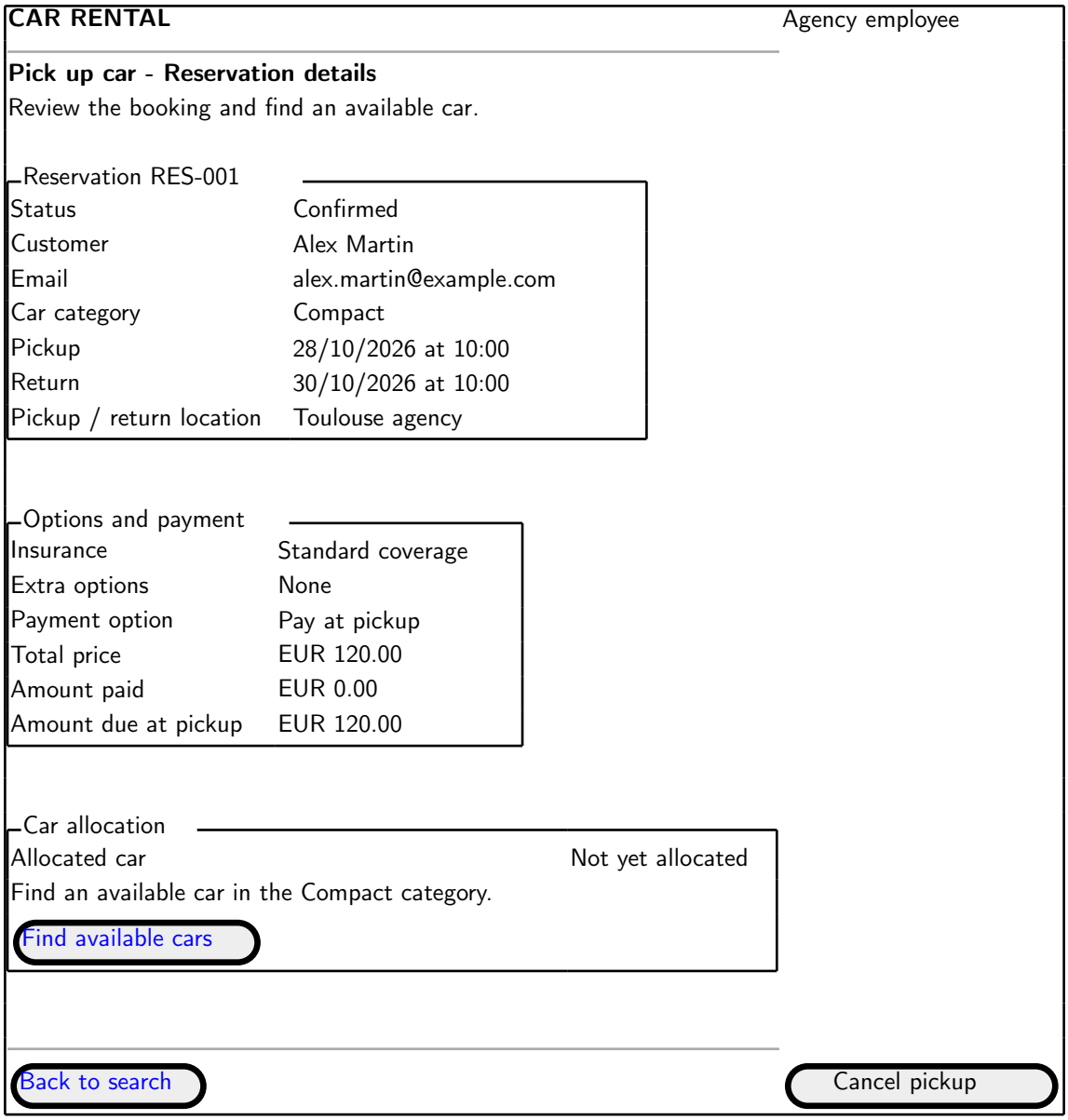


Figure 1.4: Reservation details for the pickup action

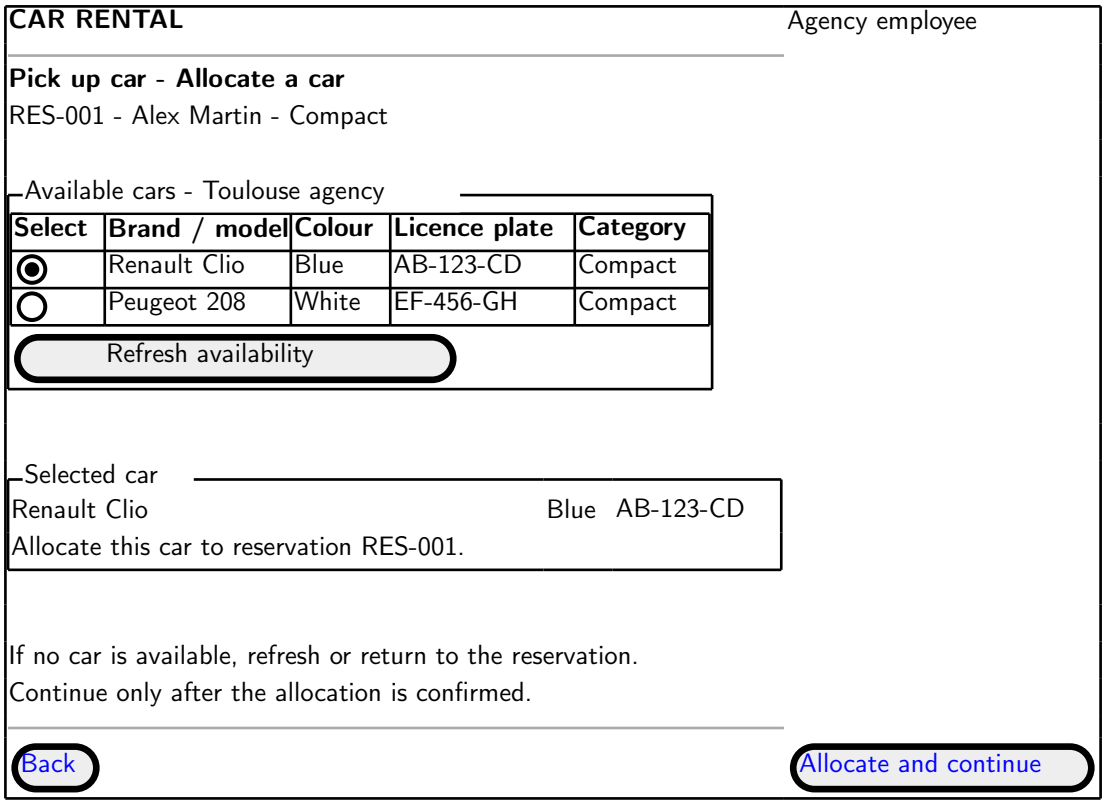


Figure 1.5: Select and allocate an available car in the booked category

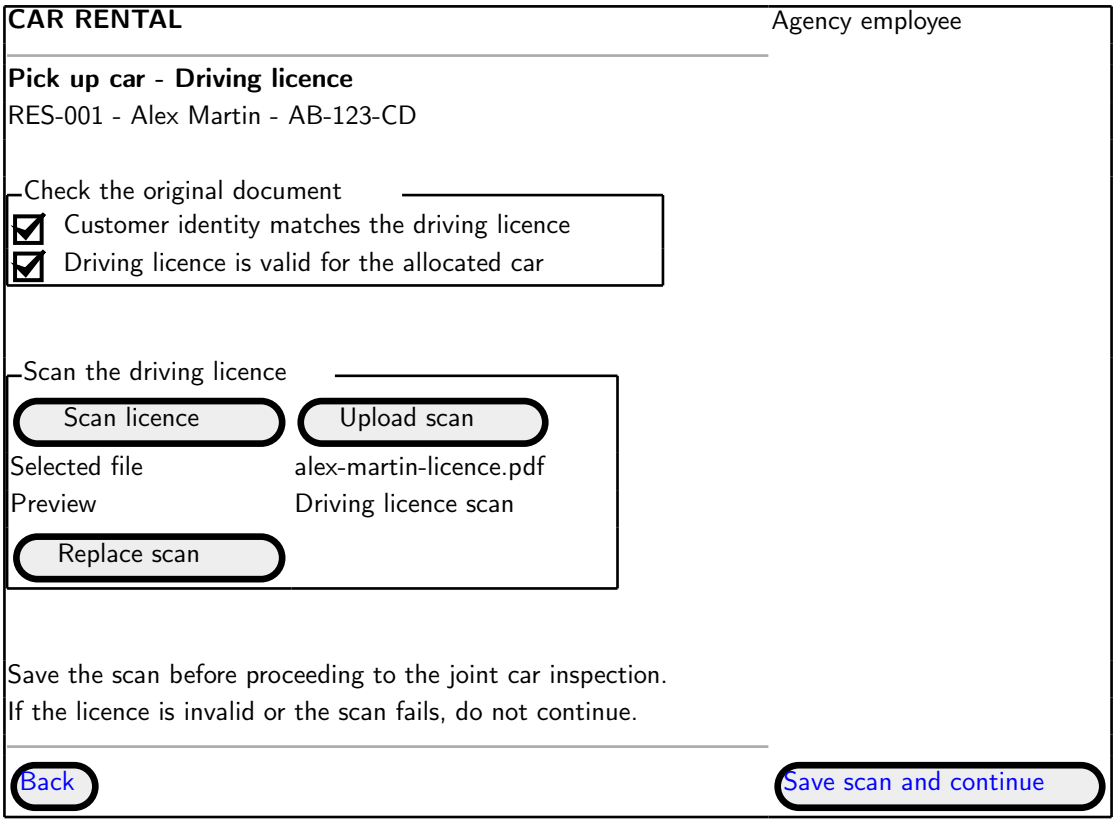


Figure 1.6: Check the driving licence and store its scan

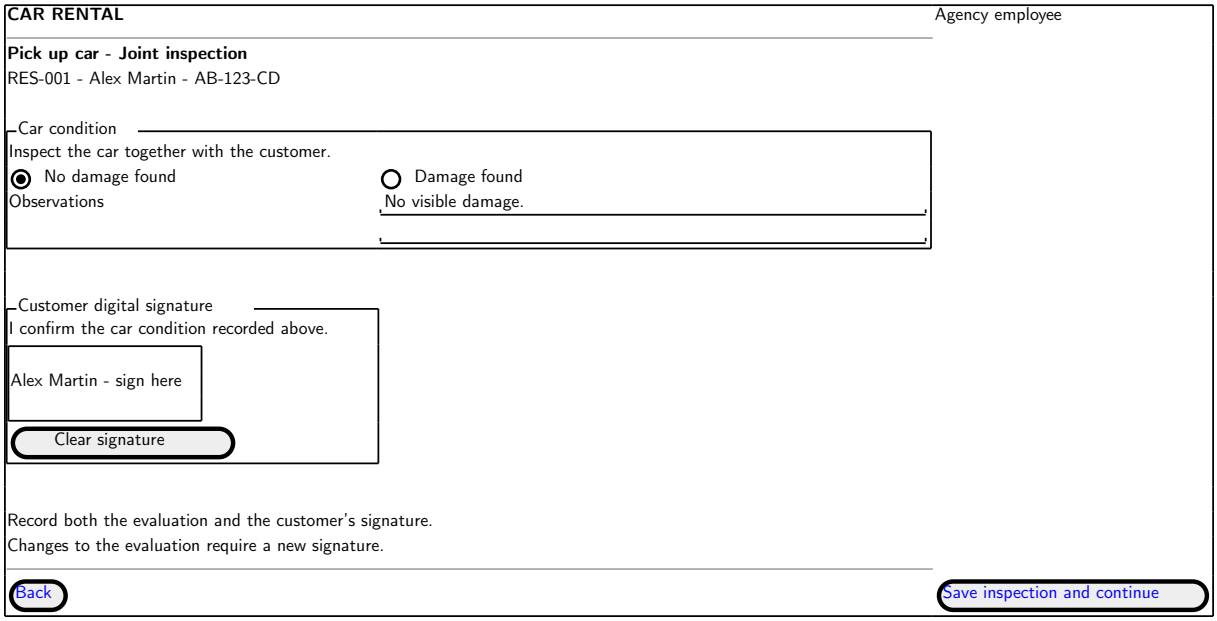


Figure 1.7: Record the joint inspection and the customer's digital signature

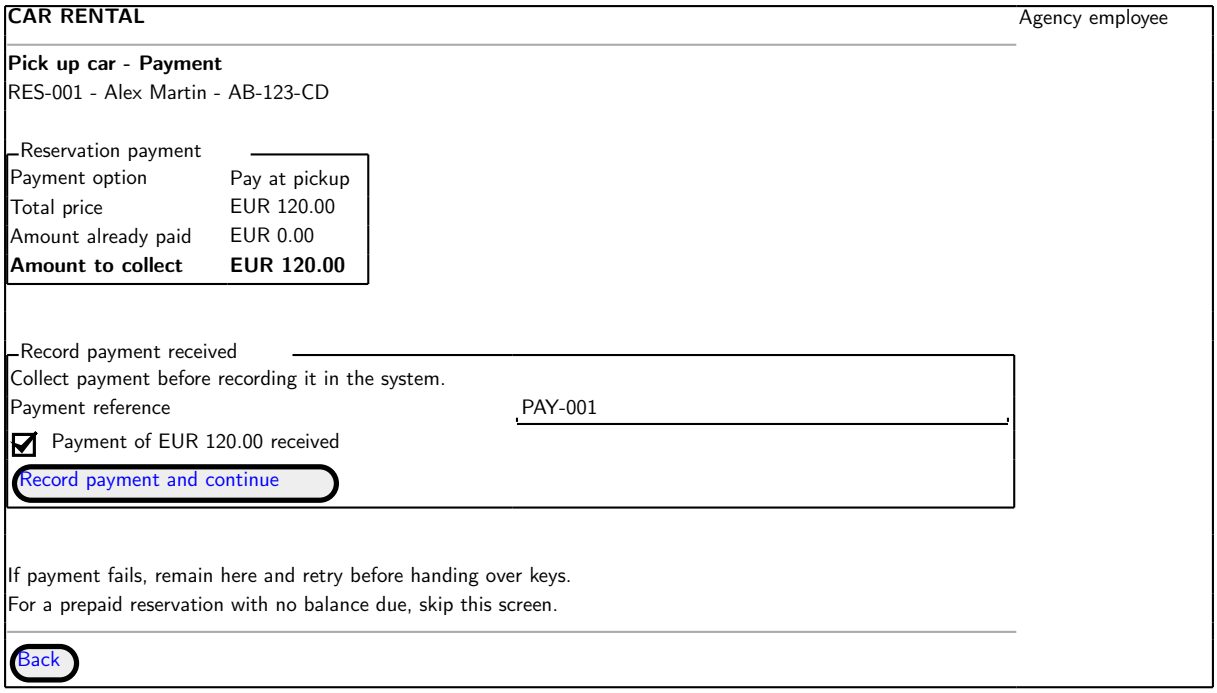


Figure 1.8: Record payment for a pay-at-pickup reservation

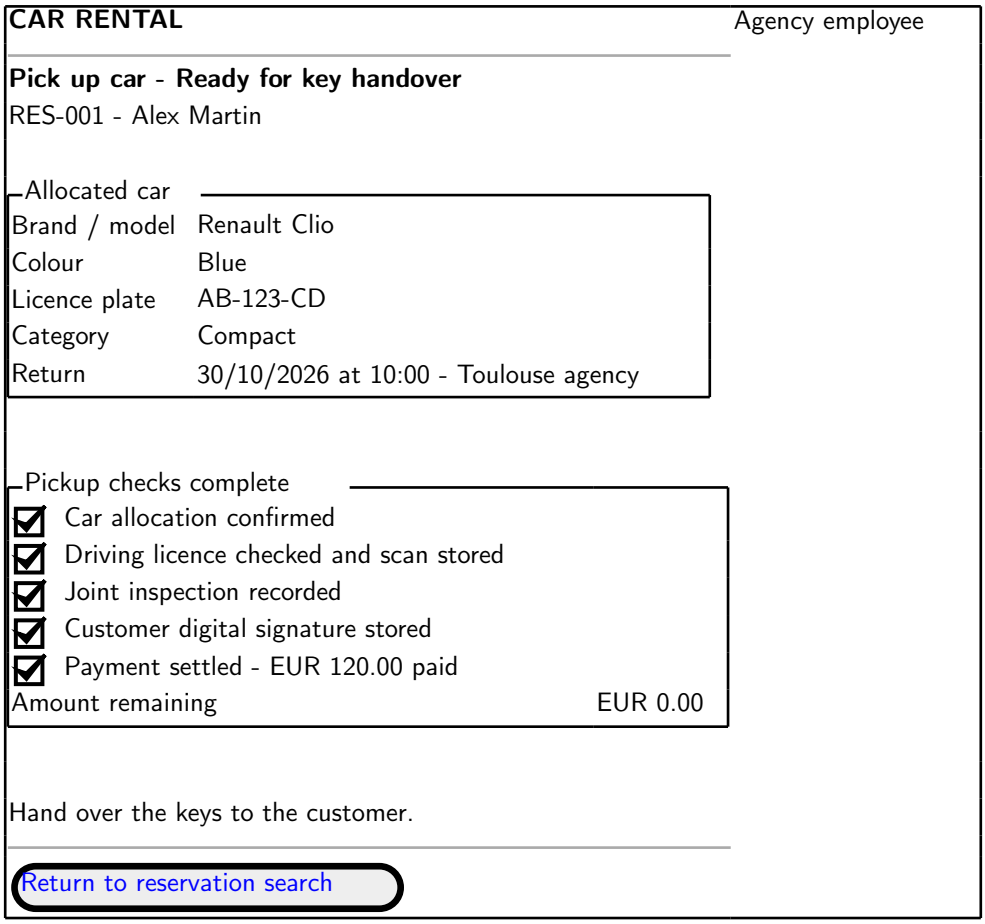


Figure 1.9: Review completed checks before handing over the keys

Chapter 2

Class Diagram

Class digram represents the main classes that is related to the primary business of the application. From the assignment text, we design the system with the following classes:

- **Customer** class represent the

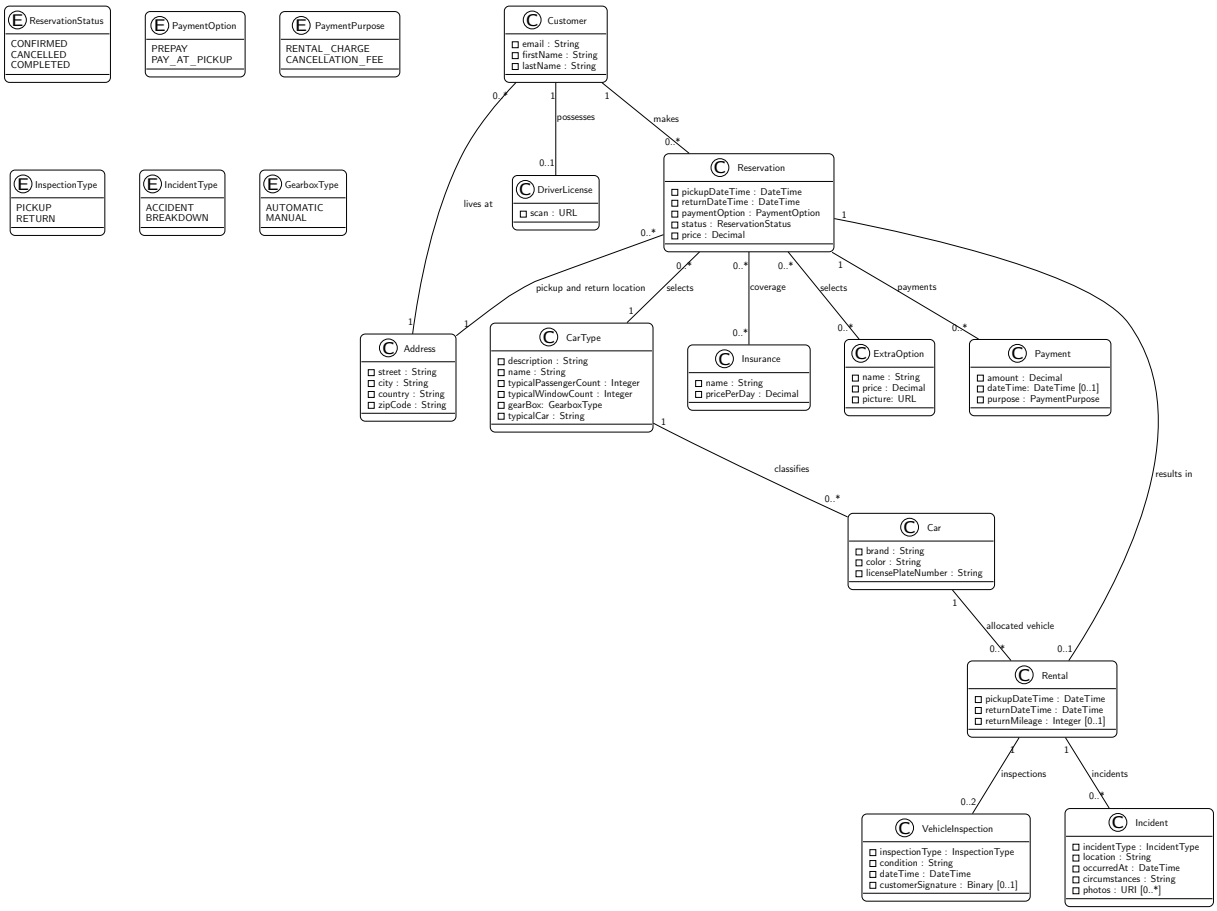


Figure 2.1: Class diagram

Chapter 3

Sequence Diagram

3.1 Reservation action

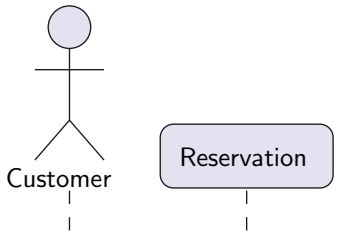


Figure 3.1: Sequence diagram for reservation